$T: P_1 \to P_2$

$T(a+bx) = x(a+bx)$

B = Basis of $P_1 = \{1, x\}$

C = " " $P_2 = \{1, x, x^2\}$

$T(a+bx) = ax + bx^2$.

$T(1) = x$ ✓

$1 = 1 + 0x$ | $x = 0 + 1x$

$a=1, b=0$ | $a=0, b=1$

$T(x) = x^2$ ✓

$[T(1)]_C = [x]_C = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$, $[T(x)]_C = [x^2]_C = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

$A = \begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}$ ✓

$T: P_2 \to P_2$

$T(p(x)) = p(3x-5)$

$B = \{1, x, x^2\}$

$T(1) = 1$

$T(x) = 3x - 5$

$T(x^2) = (3x-5)^2 = 9x^2 - 30x + 25$

$$[T(1)]_B = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix},$$

$$[T(x)]_B = \begin{pmatrix} -5 \\ 3 \\ 0 \end{pmatrix}$$

$$[T(x^2)]_B = \begin{pmatrix} 25 \\ -30 \\ 9 \end{pmatrix}$$

$$A = \begin{pmatrix} 1 & -5 & 25 \\ 0 & 3 & -30 \\ 0 & 0 & 9 \end{pmatrix}$$

$T: P_2 \to P_2$

$T(a_0 + a_1 x + a_2 x^2) = a_0 + a_1(x-1) + a_2(x-1)^2$.

$T(1) = T(1 + 0x + 0x^2) = 1 + 0(x-1) + 0(x-1)^2 = \underline{1}$

$T(x) = (x-1)$

$T(x^2) = (x-1)^2 = x^2 - 2x + 1$

$$A = \begin{pmatrix} 1 & -1 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{pmatrix}$$

$$[T(1)]_B = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad [T(x)]_B = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \quad [T(x^2)]_B = \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$$

$T: \mathbb{R}^2 \to \mathbb{R}^3$

$T\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} x_1 + 2x_2 \\ -x_1 \\ 0 \end{pmatrix}$

$B = \left\{ \begin{pmatrix} 1 \\ 3 \end{pmatrix}, \begin{pmatrix} -2 \\ 4 \end{pmatrix} \right\}$, $C = \left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 8 \\ 8 \end{pmatrix}, \begin{pmatrix} 3 \\ 0 \\ 0 \end{pmatrix} \right\}$

(1)

$T\begin{pmatrix} 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 1+2(3) \\ -1 \\ 0 \end{pmatrix} = \begin{pmatrix} 7 \\ -1 \\ 0 \end{pmatrix}$

$T\begin{pmatrix} -2 \\ 4 \end{pmatrix} = \begin{pmatrix} -2+2(4) \\ -(-2) \\ 0 \end{pmatrix} = \begin{pmatrix} 6 \\ 2 \\ 0 \end{pmatrix}$

II $\left[T\begin{pmatrix} 1 \\ 3 \end{pmatrix}\right]_C = ?$

$\left[T\begin{pmatrix} -2 \\ 4 \end{pmatrix}\right]_C = \begin{pmatrix} 0 \\ 1 \\ 4/3 \end{pmatrix}$

$c_1\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + c_2\begin{pmatrix} 2 \\ 2 \\ 0 \end{pmatrix} + c_3\begin{pmatrix} 3 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 7 \\ -1 \\ 0 \end{pmatrix}$

$\begin{pmatrix} 1 & 2 & 3 & 7 \\ 1 & 2 & 0 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix} R \begin{pmatrix} 1 & 2 & 3 & 7 \\ 0 & 0 & -3 & -8 \\ 0 & 0 & 0 & 0 \end{pmatrix}$

$c_1 = 0$, $c_2 = -\frac{1}{2}$, $c_3 = 8/3$

$A = \begin{pmatrix} 0 & 0 \\ -1/2 & 1 \\ 8/3 & 4/3 \end{pmatrix}$.

Q: Find the eigenvalues and bases for the eigenspace of the linear operator $T: P_2 \to P_2$ defined by

$$T(a+bx+cx^2) = -2c + (a+2b+c)x + (a+3c)x^2$$

Solution.

$\{1, x, x^2\}$

$T(1) = x + x^2$, $T(x) = 2x$, $T(x^2) = -2 + x + 3x^2$

$$A = \begin{pmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{pmatrix}$$

$\det(A - \lambda I) = 0 \Rightarrow \lambda = 1, 2, 2$

$$\begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} \qquad \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

Basis of eigenspace.

$E_1 = \{-2 + x + x^2\}$

$E_2 = \{-1 + x^2, \; x\}$

$T: M_{22} \to M_{22}$ ✓

$$T\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 2c & a+c \\ b-2c & d \end{pmatrix}$$

$$T\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$ ✓

$$T\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$ ✓

$$T\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ -2 & 0 \end{pmatrix}$$ ✓

$$T\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$ ✓

$$B = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\}$$

$$\left[T\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \right]_B = ? \quad \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$$ ✓

$$A = \begin{pmatrix} 0 & 0 & 2 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}_{4\times 4}$$ ✓